

PSET 1

Question 1

Let $f : X \rightarrow Y$ be a map of sets. A function $g : Y \rightarrow X$ is called a left (resp. right) inverse of f if $g \circ f = \text{id}_X$ (resp. $f \circ g = \text{id}_Y$).

- Show that f permits a right inverse if and only if f is surjective or onto. Show that in general a right inverse is not unique.
- Show that f permits a left inverse if and only if f is injective or one-to-one. Show that in general a left inverse is not unique.

Solution:

Proof. We split parts a). and b). into separate propositions:

Proposition 0.0.1

$f \circ g = \text{id}_Y \iff f$ is surjective. In general id_Y , the right identity function, is not unique.

Let f be a surjective function. For all y , $\exists x$, such that $f(x) = y$; we define $g(y)$ to be such an x . Then, since $g(y) = x$, we have $f(x) = f(g(y))$ but $f(x) = y$ thus $y = f(g(y)) \forall y$ which is the definition of our identity function. Thus we have shown surjectivity implies $f \circ g = \text{id}_Y$. Now let $f \circ g = \text{id}_Y$, then since $\forall y$, $g(y) = x$, we have $f(g(y)) = f(x) \implies y = f(x) \forall y$. Thus we have shown $f \circ g = \text{id}_Y \implies$ surjectivity. Since both cases are satisfied the proposition is true. In general, the right inverse is not unique—consider $f : X \rightarrow Y$ such that $X = \mathbb{N}$ and $Y = \{0\}$; there exist infinitely many right inverses since any $x \in X$ can map to the only $y \in Y$.

Proposition 0.0.2

$g \circ f = \text{id}_X \iff f$ is injective. In general id_X , the left identity function, is not unique.

Let f be an injective function. Let there be a $g : Y \rightarrow X$ such that $g(y) = x$. Then for some $y_1 = f(x_1)$, let there be a $g(y_n) = x_n$ given by $g : Y \rightarrow X$. Note that $g(y)$ can only have list of x s equal to the cardinality of the image of f ; then we have $g(y_1) = g(f(x_1))$, note that $g(y_1) = x_1$, thus $x_1 = g(f(x_1)) \forall x \in X$, which means that $g(f(x)) = g \circ f = \text{id}_X$. Thus injectivity $\implies g \circ f = \text{id}_X$. Now let $g \circ f = \text{id}_X$. Let there exist a pair of functions such that $f(x_1) = f(x_2)$. We show that $\forall f(x_1) = f(x_2) \implies x_1 = x_2$. Given $f(x_1) = f(x_2) \implies g(f(x_1)) = g(f(x_2))$, but $g(f(x)) = \text{id}_X$, thus $x_1 = x_2$. So $g \circ f \implies$ injectivity. To show that a left inverse lacks uniqueness consider $f : \{0, 1\} \hookrightarrow \{a, b, c\}$, by pigeonhole principle, at least one $g(n)$ must map to the same number as another $g(n)$, thus our $g(y) = \text{id}_X$ is not unique. $\square$

Question 2

Solution: